CHITTAGONG UNIVERSITY OF ENGINEERING AND TECHNOLOGY

DEPARTMENT OF BIOMEDICAL ENGINEERING

BME 416 - PROJECT REPORT

Comprehensive Multi-Zonal Adaptive Thermo-regulation and Air Quality Optimization System for Healthcare Facility Environmental Control

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Date: November 11, 2024

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1 Introduction

1.1 Background

Effective environmental control in healthcare facilities has been a focal point in studies exploring the use of IoT solutions to automate climate regulation. Existing research highlights the importance of temperature and humidity regulation in minimizing infection rates and ensuring patient comfort. For example, studies show that patient health outcomes improve significantly in facilities where precise environmental control mechanisms are deployed, reducing recovery time and enhancing overall patient satisfaction. Recent advances in IoT technology, including wireless sensors and cloud platforms, enable real-time monitoring and data-driven environmental control. Systems integrating components like the DHT11 temperature and humidity sensor and microcontrollers such as the ESP8266 have proven effective in monitoring environmental variables. The Blynk platform is widely utilized for remote data access, offering healthcare personnel the ability to monitor conditions from mobile devices and computers. Despite these advancements, there remains a need for systems that can automatically respond to changing environmental parameters, providing immediate alerts and taking corrective action. This project builds on these studies by developing an adaptive solution specifically tailored to healthcare facilities, with a focus on thermal regulation and the potential for future expansion to air quality monitoring.

1.2 Problem Statement

Healthcare facilities face critical challenges in maintaining safe and comfortable environmental conditions across different zones. Fluctuating temperatures and humidity levels can negatively impact patient health, increase the risk of equipment malfunction, and compromise overall facility operations. Given the specific needs for consistent climate control, especially in sensitive zones such as patient wards and operating rooms, a reliable and adaptable system is essential. Traditional manual monitoring and control systems may be insufficient for dynamic, multi-zonal environments, where real-time data and automated responses are crucial for timely adjustments. This project seeks to address these issues by developing an IoT-based, adaptive thermoregulation and air quality monitoring system that provides continuous, real-time monitoring and control, ensuring optimal environmental conditions in healthcare settings.

2 Methodology

The system design includes:

1. **Data Collection:** A DHT11 sensor is used in each zone to measure temperature and humidity levels continuously.

2. **Wireless Data Transmission:** Each DHT11 sensor transmits data to an ESP8266 microcontroller, which sends the information to the Blynk platform for remote monitoring.

3. **Adaptive Response Mechanism:** An LED indicator provides an immediate visual alert if the temperature exceeds a 31°C threshold, signaling potential thermal

discomfort. The system is configured to turn the LED off when the temperature decreases to 30°C or lower. This control mechanism is adaptable across zones, allowing healthcare personnel to quickly identify and address areas with rising temperatures.

4. **Air Quality Control Integration (Future Expansion):** While not yet implemented, additional sensors could be incorporated to monitor air quality metrics, such as CO2 levels, to provide a more comprehensive environmental control solution.

3 Component Integration

The project components are selected for their compatibility and functionality in healthcare environments:

1. **DHT11 Sensor:** Provides accurate temperature and humidity readings in each zone.
2. **ESP8266 WiFi Module:** Acts as the primary processing unit, receiving data from sensors and controlling the relay and LED as needed.
3. **Relay Module and LED:** The relay controls the LED, which visually signals elevated temperatures, providing immediate, zone-specific feedback.
4. **Blynk Platform:** Displays real-time environmental data across zones, accessible via mobile and laptop, enhancing staff ability to monitor conditions from anywhere within the facility.

4 Development of the Adaptive Thermoregulation Algorithm

Flowchart of an adaptive thermoregulation algorithm controls the LED based on temperature thresholds:

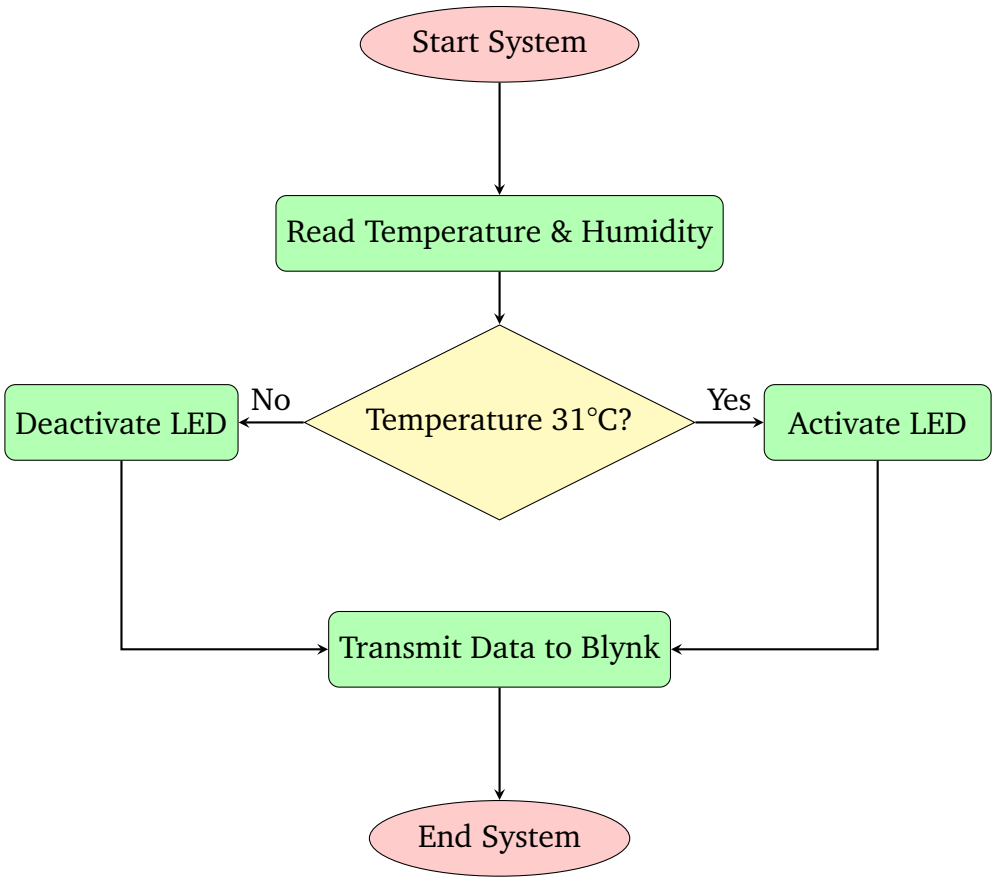

Figure 1: Enhanced Flowchart of Adaptive Thermoregulation Algorithm

5 Results and Analysis

Testing in multiple zones showed effective real-time data capture and consistent LED response to temperature fluctuations:

1. ***Accuracy:*** The DHT11 provided temperature readings with an acceptable range of accuracy for healthcare requirements.

2. ***Response Time:*** The ESP8266 and relay setup provided rapid LED activation when the threshold was crossed, allowing prompt identification of high-temperature zones.

3. ***Remote Monitoring:*** The Blynk platform displayed data with minimal delay, ensuring reliable, real-time remote monitoring capability for staff. These results demonstrate the system's potential for effective environmental control in healthcare facilities, where immediate response to temperature changes is critical.

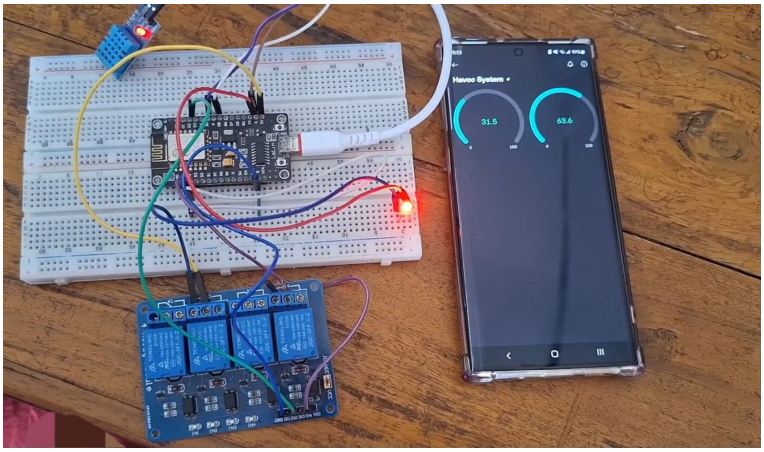

Figure 2:Hardware setup

6 Conclusion

This project successfully demonstrates an IoT-based, multi-zonal solution for adaptive thermoregulation and potential air quality monitoring in healthcare facilities. By integrating environmental sensors with wireless data transmission and adaptive algorithms, the system enhances patient safety and operational efficiency. With its scalability, it can be expanded to include additional environmental parameters, such as air quality indices, making it a versatile tool for healthcare facility management.
Future Scope: Further development could involve adding sensors for CO2 and particulate matter to create a fully integrated air quality control system. Additionally, machine learning models could be introduced to dynamically adjust threshold levels based on historical data, enabling proactive environmental management.

7 Necessary code

```
define BLYNK_TEMPLATE_ID "TMPL6F3xaax5I"

define BLYNK_TEMPLATE_NAME "HavocSystem"
define BLYNK_AUTH_TOKEN "PBlOOuaeaZMNvc3vYI_a9Mkvqm7QUGOK"
define BLYNK_PRINT Serial
include <ESP8266WiFi.h>
include <BlynkSimpleEsp8266.h>
include <DHT.h>

define DHTPIN D5 // Digital pin connected to the DHT sensor

define DHTTYPE DHT11 // DHT 11
define FAN_PIND2//FANRELAY

char auth[] = BLYNK_AUTH_TOKEN; char ssid[] = "CUET--Students"; //Replace with your WiFi SSID
char pass[] = "1020304050"; //Replace with your WiFi password

float humDHT = 0;
float tempDHT = 0;

// Initialize DHT sensor.
DHT dht(DHTPIN, DHTTYPE); //communicate with sensor

void setup()
Serial.begin(115200); // Set the baud rate
pinMode(FAN_PIN, OUTPUT);
digitalWrite(FAN_PIN, LOW);

Serial.println(F("DHTxx test!")); // Debug message

dht.begin(); // Start DHT sensor
Serial.println("DHT sensor initialized"); // Debug message

Serial.println("Connecting to WiFi...");
Blynk.begin(auth, ssid, pass); // Connect to Blynk
Serial.println("Connected to WiFi!"); // Debug message

Serial.println("Setup complete!"); // Confirmation that setup is done


void loop()
Blynk.run();

// Wait a few seconds between measurements.
delay(2000);
```

```
// Reading temperature or humidity takes about 250 milliseconds!
humDHT = dht.readHumidity();
tempDHT = dht.readTemperature(); // Read temperature in Celsius

// Check if any reads failed and exit early (to try again).
if (isnan(humDHT) —— isnan(tempDHT))
Serial.println("Failed to read from DHT sensor!");
return;


// Output the temperature and humidity readings to the serial monitor.
Serial.print(F("Temperature: "));
Serial.print(tempDHT);
Serial.println(F("°C"));

Serial.print(F("Humidity: "));
Serial.print(humDHT);
Serial.println(F("
Serial.println("*");
Serial.println();

// Control the fan based on temperature
if (tempDHT ¿ 31)
digitalWrite(FAN_PIN,HIGH);//Turnthefanon
Serial.println("FanON");//Debugmessage
elsedigitalWrite(FAN_PIN,LOW);//TurnthefanoffSerial.println("FanOFF");//Debugmessage

// Send temperature and humidity to Blynk app
Blynk.virtualWrite(V1, tempDHT);
Blynk.virtualWrite(V2, humDHT);
```